10

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CHE 313
HOMEWORK #2

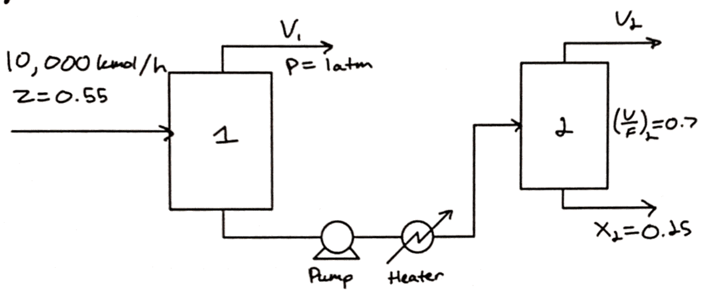
1) PROBLEM 2.D2

Two flash distillation chambers are hooked together as shown in the diagram. Both are at 1.0 atm pressure. The feed to the first drum is a binary mixture of methanol and water that is 55.0 mol% methanol. Feed flow rate is 10,000 kmol/h. The second flash drum operates with $(V/F)_2 = 0.7$ and the liquid product composition is 25.0 mol% methanol. Equilibrium data are given in table 2-7.

TABLE 2-7. *Vapor-liquid equilibrium data for methanol water (p = 1 atm) (mol%)*

Methanol Liquid	Methanol Vapor	Temp., °C
0	0	100
2.0	13.4	96.4
4.0	23.0	93.5
6.0	30.4	91.2
8.0	36.5	89.3
10.0	41.8	87.7
15.0	51.7	84.4
20.0	57.9	81.7
30.0	66.5	78.0
40.0	72.9	75.3
50.0	77.9	73.1
60.0	82.5	71.2
70.0	87.0	69.3
80.0	91.5	67.6
90.0	95.8	66.0
95.0	97.9	65.0
100.0	100.0	64.5

Source: Perry et al. (1963), p. 13-5.



A) What is the fraction vaporized in the first flash drum?

Step1. Using the known methanol fraction in the liquid leaving Drum #2, we can determine the vapor fraction of methanol w/ the data provided

$X_2 = 0.25 \therefore Y_2 = \frac{(0.579 + 0.665)}{2} = 0.622$

Step 2: Using (X_2, Y_2) & the $(\frac{V}{F})_2 = 0.7$ we can determine the composition of the feed.

$$y = -\frac{L}{V}x + \frac{F}{V}z \Rightarrow y = -\left(\frac{F-V}{V}\right)x + \frac{F}{V}z \Rightarrow y = \left(-\frac{F}{V}+1\right)x + \left(\frac{F}{V}\right)z$$

$$\Rightarrow \left(\frac{F}{V}\right)z = y - \left(-\frac{F}{V}+1\right)x \Rightarrow \underbrace{z = \frac{V}{F}\left(y - \left(-\frac{F}{V}+1\right)x\right)}_{\text{plug in values}}$$

$$z = 0.7\left(0.622 - \left(\frac{-1}{0.7}+1\right)0.25\right) = 0.51 \qquad Z_2 = 0.51$$

Step3: Since $(Z_2 = 0.51)$ is the composition of the feed into drum #2, it can be used as the X_1 fraction since the feed into drum #2 is the liquid stream leaving drum #1. Using the $X_1, Y_1, F_1, \& Z_1$ we can solve for the $(V/F)_1$.

$F = 10{,}000 \frac{kmol}{h}$

$X_1 = 0.51 \therefore Y_1 = 0.78$ → FROM ~~DATA~~ (used graph attached below)

$z = 0.55$
(X_1, Y_1) slope: $\frac{0.78 - 0.55}{0.51 - 0.55} = -5.75$, $\quad y = -\frac{L}{V}x + \underbrace{\frac{F}{V}z}$ solve for using slope & points

$0.55 = -5.75(0.55) + b$

$b = 3.7125$

$b = \frac{F}{V}z \Rightarrow \left(\frac{V}{F}\right)_1 = \frac{z}{b} = \frac{0.55}{3.7125}$

$$\boxed{\left(\frac{V}{F}\right)_1 = 0.148}$$

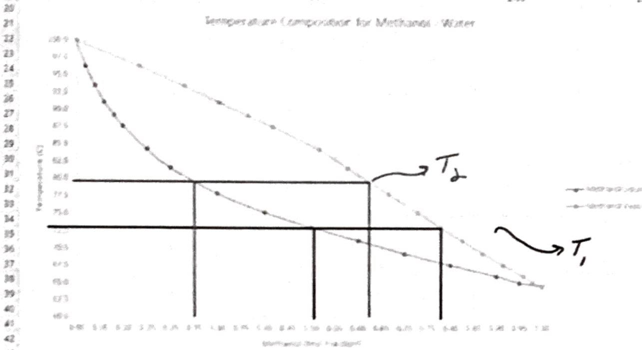
B) What are $Y_1, Y_2, X_1, T_1, \& T_2$?

$X_1 = 0.51$
$Y_1 = 0.78$
$Y_2 = 0.62$
Determined above

Flash drum #1 operating line
(Determined above)
$\{y = -5.75x + 3.7125\}$

Temperature Composition Diagram for Methanol Water

Methanol Liquid (mol%)	Methanol Vapor (mol%)	Temperature	Pressure (atm)	Methanol Liquid (Mol Frac)	Methanol Vapor (Mol Frac)
0.00	0.00	100.00	1.00	0.00	0.00
2.00	13.40	96.40		0.02	0.13
4.00	23.00	93.50		0.04	0.23
6.00	30.40	91.20		0.06	0.30
8.00	36.50	89.30		0.08	0.37
10.00	41.80	87.70		0.10	0.42
15.00	51.70	84.40		0.15	0.52
20.00	57.90	81.70		0.20	0.58
30.00	66.50	78.00		0.30	0.67
40.00	72.90	75.30		0.40	0.73
50.00	77.90	73.10		0.50	0.78
60.00	82.50	71.20		0.60	0.83
70.00	87.00	69.30		0.75	0.87
80.00	91.50	67.60		0.80	0.92
90.00	95.80	66.00		0.90	0.96
95.00	97.90	65.00		0.95	0.98
100.00	100.00	64.50		1.00	1.00



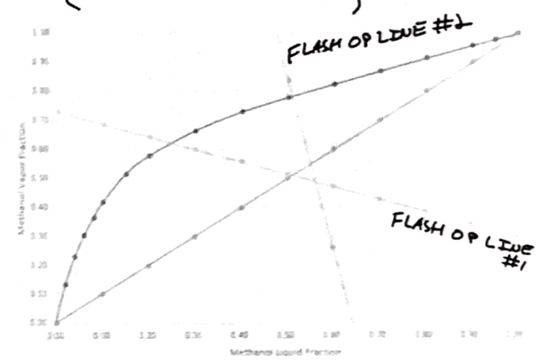
Finding Flash Drum #2 Operating line

$z = 0.51 \quad \left(\frac{V}{F}\right)_2 = 0.7$

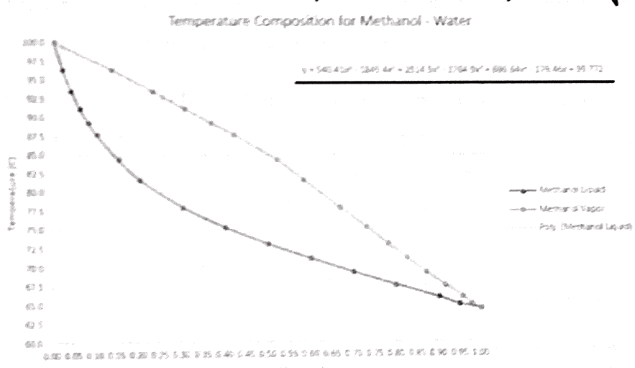
$$y = -\frac{L}{V}x + \frac{F}{V}z \Rightarrow y = -\left(\frac{F}{V} - 1\right)x + \frac{F}{V}z$$

$$y = -\left(\frac{1}{0.7} - 1\right)x + \frac{1}{0.7}(0.51)$$

$$\left\{ y = -0.429x + 0.729 \right\}$$



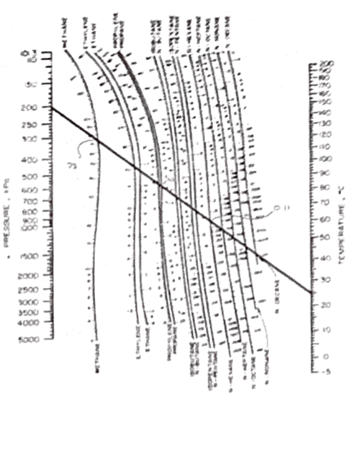
For more exact temperature values, Excel generated 6th order polynomial



$$\boxed{\begin{aligned} T_1 &= 72.8^\circ C \\ T_2 &= 79.8^\circ C \\ X_1 &= 0.51 \\ Y_1 &= 0.78 \\ Y_2 &= 0.62 \end{aligned}}$$

2) PROBLEM 2 D8

You want to flash a mixture w/ a drum of 2.0 atm and a drum temperature of 25°C. The feed is 2000.0 kmol/h. The feed is 5.0 mol% methane, 10.0 mol% propane, and the rest n-hexane. Find the fraction vaporized, vapor mole fractions, liquid mole fractions, and vapor & liquid flow rates. Use DePriester charts.



$$P = 2.0\,\text{atm} \times \frac{101.325\,\text{kPa}}{1\,\text{atm}} = 202.65\,\text{kPa}$$

STEP 1: FIND THE K VALUES FROM THE CHART

#1 Methane : $k_1 = 73$, $z_1 = 0.05$

#2 Propane : $k_2 = 4.1$, $z_2 = 0.10$

#2 n-hexane : $k_3 = 0.11$, $z_3 = 0.85$

STEP 2: SOLVE RR & FIND (V/F)

Rachford Rice Equation

$$\sum \frac{(k_i - 1) z_i}{1 + (k_i - 1) V/F} = 0$$

$$\left[\frac{(73-1)(0.05)}{1+(73-1)\left(\frac{V}{F}\right)}+\frac{(4.1-1)(0.10)}{1+(4.1-1)\left(\frac{V}{F}\right)}+\frac{(0.11-1)(0.85)}{1+(0.11-1)\left(\frac{V}{F}\right)}=0\right]$$

$$\left[\frac{\left(\frac{18}{5}\right)}{1+72\left(\frac{V}{F}\right)}+\frac{\left(\frac{31}{100}\right)}{1+\frac{31}{10}\left(\frac{V}{F}\right)}+\frac{\frac{-1513}{2000}}{1-\frac{89}{100}\left(\frac{V}{F}\right)}=0\right]$$

$$\left(\frac{18}{5}\right)\left(1+\frac{31}{10}\left(\frac{V}{F}\right)\right)\left(1-\frac{89}{100}\left(\frac{V}{F}\right)\right)+\left(\frac{31}{100}\right)\left(1+72\left(\frac{V}{F}\right)\right)\left(1-\frac{89}{100}\left(\frac{V}{F}\right)\right)-\left(\frac{1513}{2000}\right)\left(1+72\left(\frac{V}{F}\right)\right)\left(1+\frac{31}{10}\left(\frac{V}{F}\right)\right)=0$$

$$\left(\frac{18}{5}\right)\left(1-\frac{89}{100}\left(\frac{V}{F}\right)+\frac{31}{10}\left(\frac{V}{F}\right)-\frac{2759}{1000}\left(\frac{V}{F}\right)^2\right)\Rightarrow\left(\frac{18}{5}+\frac{1989}{250}\left(\frac{V}{F}\right)-\frac{24831}{2500}\left(\frac{V}{F}\right)^2\right)$$

$$\left(\frac{31}{100}\right)\left(1-\frac{89}{100}\left(\frac{V}{F}\right)+72\left(\frac{V}{F}\right)-\frac{1602}{25}\left(\frac{V}{F}\right)^2\right)\Rightarrow\left(\frac{31}{100}+\frac{220441}{10000}\left(\frac{V}{F}\right)-\frac{24831}{1250}\left(\frac{V}{F}\right)^2\right)$$

$$\left(-\frac{1513}{2000}\right)\left(1+\frac{31}{10}\left(\frac{V}{F}\right)+72\left(\frac{V}{F}\right)+\frac{1116}{5}\left(\frac{V}{F}\right)^2\right)\Rightarrow\left(\frac{-1513}{2000}-\frac{1136263}{20000}\left(\frac{V}{F}\right)-\frac{422127}{2500}\left(\frac{V}{F}\right)^2\right)$$

$$-\frac{24831}{125}\left(\frac{V}{F}\right)^2-\frac{536261}{20000}\left(\frac{V}{F}\right)+\frac{6307}{2000}=0$$

$$\boxed{\frac{V}{F}=0.075}$$

$$\frac{V}{F}=\frac{\frac{536261}{20000}\pm\sqrt{\left(\frac{536261}{20000}\right)^2-4\left(-\frac{24831}{125}\right)\left(\frac{6307}{2000}\right)}}{2\left(-\frac{24831}{125}\right)}=0.0754 \text{ or } -0.2104$$

STEP 3: FIND FLOWRATES USING BALANCE

Using $\frac{V}{F}=0.075$ & $F=2000.0$ kmol/h, & w/ $F=V+L$, $L=F-V$ ∴

$V=150.89$ kmol/h
$L=1849.11$ kmol/h

STEP 4: FIND LIQUID MOLE FRACTIONS

$$x_i=\frac{z_i}{1+(k_i-1)\frac{V}{F}} : x_1=\frac{0.05}{1+(73-1)(0.075)}=0.0078,\quad y_1=73(0.0078)=0.5675$$

$$y_i=k_ix_i \qquad x_2=\frac{0.10}{1+(4.1-1)(0.075)}=0.0810,\quad y_2=4.1(0.0810)=0.3323$$

$$x_3=\frac{0.85}{1+(0.11-1)(0.75)}=0.9112,\quad y_3=0.11(0.9112)=0.1002$$

$x_1=0.0078$	$y_1=0.5675$
$x_2=0.0810$	$y_2=0.3323$
$x_3=0.9112$	$y_3=0.1002$

3) PROBLEM 2.D9

We wish to flash distil an ethanol water mixture that is 30.0 wt% ethanol and has a feed flow of 1000.0 kg/h. Feed is at 200 C°. The flash drum operates at a pressure of 10 kg/cm². Find T_{drum}, weight fraction of liquid & vapor products, and liquid & vapor flowrates.

DATA:

$C_{PL,EtOH}$: 37.96 at 100°C (kcal/kmol°C)

$C_{PL,W}$: 18.0, (kcal/kmol·C)

$C_{PV,EtOH}$: $14.66+3.758\times10^{-2}T-2.091\times10^{-5}T^2+4.74\times10^{-9}T^3$

$C_{PV,W}$: $7.88+0.32\times10^{-2}T-0.04833\times10^{-5}T^2$

→ Both are in units of (kcal/kmol°C), & T is in °C

ρ_{EtOH}: 0.789 g/mL

ρ_W: 1.0 g/mL

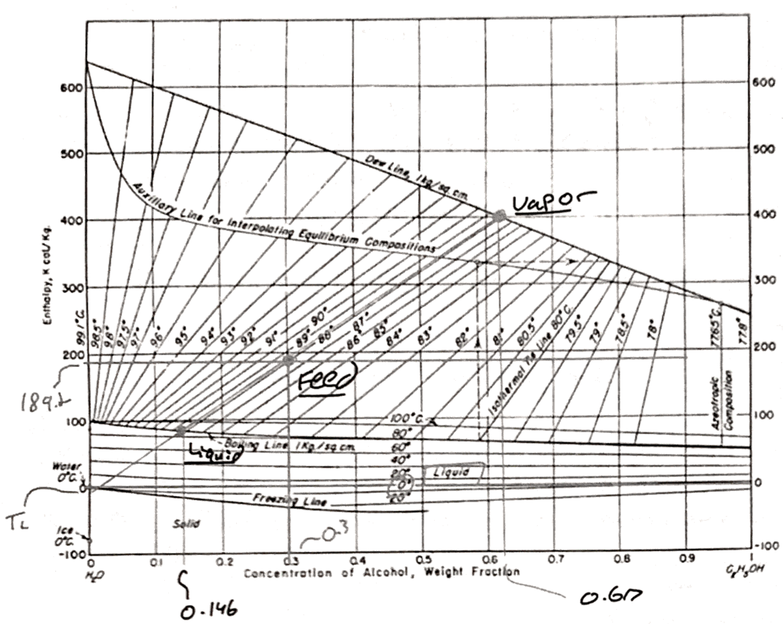
MW_{EtOH}: 46.07

MW_W: 18.016

λ_{EtOH}: 9.22 $\left(\frac{kcal}{mol}\right)$ at 351.7 K

λ_W: 9.7171 $\left(\frac{kcal}{mol}\right)$ at 373.16 K

Enthalpy Composition diagram at $p = 1\,kg/cm^2$



$$P = \frac{1.0\,kg}{cm^2} \times \frac{9.81\,m/s^2}{\left(\frac{1\,m^2}{(100)^2 cm^2}\right)} = 98100\,\frac{N}{m^2}$$

$\Rightarrow 98100\,Pa \Rightarrow 98.1\,kPa = 0.968\,atm$

STEP#1: Plot the line for the liquid using data provided.

$$h_F = C_{PL}(T_F - T_L)$$

Mixture — feed temp — liquid reference (From chart $T_L = 0°C$)

Finding C_{PL}

$$C_{PL} = X_w C_{PL,w} + X_{EtOH} C_{PL,EtOH}$$

Using wt % fractions

30% wt ethanol & 70% wt water $\Rightarrow X_w = 70\,kg\,water \times \frac{1}{MW_{water}} = \frac{70\,kg}{18.016\,\frac{kg}{kmol}} = 3.885\,kmol\ water$

* assume 100 kg

$X_{EtOH} = \frac{Mass\,EtOH}{MW_{EtOH}} = \frac{30\,kg}{46.07\,\frac{kg}{kmol}} = 0.651\,kmol\ of\ EtOH$

Divided by Total moles

$X_w = 0.8565$

$X_{EtOH} = 0.1435$

Now back to finding C_{PL}

$$C_{PL} = X_w C_{PL,w} + X_{EtOH} C_{PL,EtOH}$$

$$= (0.8565)\left(18.0\,\frac{kcal}{kmol\,°C}\right) + (0.1435)\left(37.96\,\frac{kcal}{kmol\,°C}\right) = 20.8643\,\frac{kcal}{kmol\,°C}$$

$C_{PL} = 20.8643\,\frac{kcal}{kmol\,°C}$ since C_{PL} is a mixture we use

$MW = MW_w(X_w) + MW_{EtOH}(X_{EtOH})$

$MW = (18.016)(0.8565) + (46.07)(0.1435)$

$MW = 22.0417\,\frac{kg}{kmol}$

$h_L = C_{PL}(T_F - T_L)$

Need in $\frac{kcal}{kg}$ ∴ convert?

$$C_{PL}\left(\frac{kcal}{kg\,°C}\right) = C_{PL}\left(\frac{kcal}{kmol\,°C}\right)\cdot\left(\frac{1}{MW}\right)$$

$$= 20.8643\,\frac{kcal}{kmol\,°C} \times \left(\frac{kmol}{22.0417\,kg}\right)$$

$$C_{PL} = 0.946581\,\frac{kcal}{kg\,°C}$$

use this C_{PL} corrected units to match figure

$$h_F = \left(0.946581\,\frac{kcal}{kg\,°C}\right)(200°C - 0°C)$$

$$h_F = 189.316\,\frac{kcal}{kg}$$

STEP 2: Using h_L, & $X_F(wt\%) = 0.3$, we can plot a point on the figure and estimate the temperature of the drum. Then by drawing a line for the temperature "following trend" we can find X_{EtOH} & Y_{EtOH} in the weight percents.

$T_{drum} = 88.3°C$ $X_{EtOH} = 0.146$ wt frac. & $Y_{EtOH} = 0.617$ wt frac.

$X_w = 1 - X_{EtOH}$ & $Y_w = 1 - Y_{EtOH}$

STEP3: Find V & L using $F = V + L$

$F = V + L$

$Fz = Vy + Lx$ each component

$F = 1000.0\ \frac{kg}{h}$

So use wt fraction

(1) ETOH: $(1000)(0.30) = V(0.617) + L(0.146)$

(2) Water: $(1000)(0.70) = V(0.383) + L(0.854)$

Two equations w/ two unknowns

(1) $300 = 0.617V + 0.146L$

$V = \frac{300 - 0.146L}{0.617}$

plug into

(2) $700 = \left(\frac{300 - 0.146L}{0.617}\right)(0.383) + 0.854L$

$L = 673.1$ kg/h

$F = V + L \Rightarrow V = 326.9$ kg/h

$T_{Drum} = 88.3\,^\circ C$

$X_{ETOH} = 0.146$ (wt fraction)

$Y_{ETOH} = 0.617$ (wt fraction)

$X_{water} = 0.854$ (wt fraction)

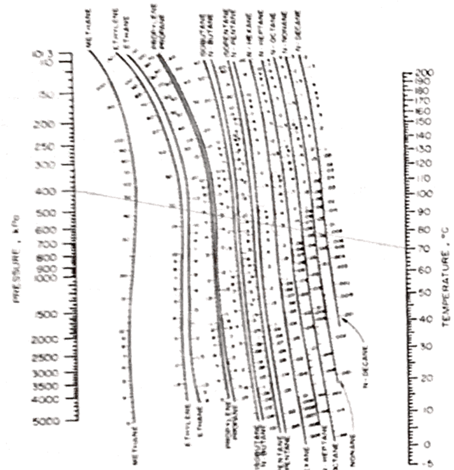
$Y_{water} = 0.383$ (wt fraction)

$V = 326.9$ kg/h

$L = 673.1$ kg/h

4) PROBLEM 2.D10

We have a mixture that is 35.0 mol % n-butane w/ unknown amounts of propane and n-hexane. We are able to operate a flash drum at 400 kPa and 70 °C w/ $x_{C6} = 0.7$. Find the mole fraction of n-hexane in the feed, z_{C6}, and the value V/F.



STEP1: Finding the k values for the components at the specified temp & pressure (use DePriester Chart)

Propane: $k_1 = 5.0$ $\quad z_1 = z_1$

n-Butane: $k_2 = 1.9$ $\quad z_2 = 0.35$

n-Hexane: $k_3 = 0.3$ $\quad z_3 = (1 - z_2 - z_1) = 0.65 - z_1$

STEP2: Set up equations and unknowns

Equations

$$x_i = \frac{z_i}{1 + (k_i - 1)\frac{V}{F}}, \quad y_i = k_i x_i, \quad \sum x_i = 1, \quad \sum y_i = 1, \quad \sum z_i = 1$$

Rachford-Rice

$$\sum \frac{(k_i - 1) z_i}{1 + (k_i - 1) V/F} = 0$$

$$x_3 = \frac{z_3}{1 + (0.3 - 1)\frac{V}{F}} \Rightarrow x_3 = \underbrace{\frac{0.65 - z_1}{1 - 0.7\left(\frac{V}{F}\right)}}_{\text{equation \#1}}$$

$$\left\{ 0.7 = \frac{0.65 - z_1}{1 - 0.7\left(\frac{V}{F}\right)} \right\}$$

Rachford-Rice

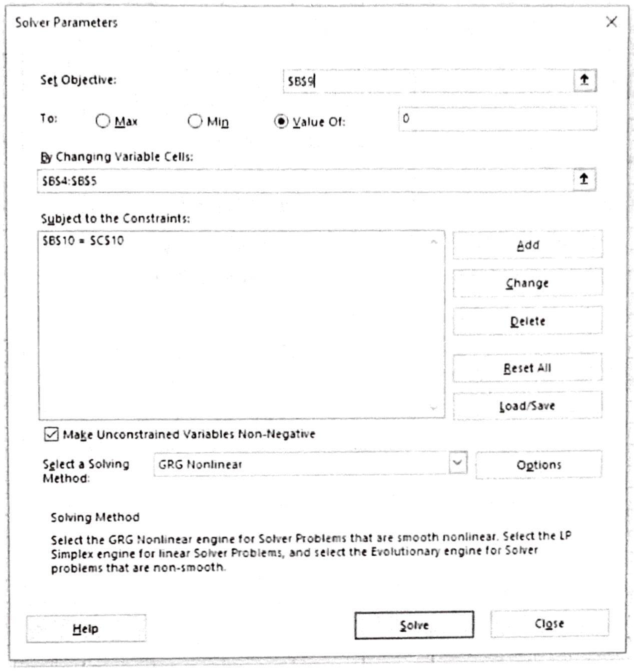
$$\frac{(k_1 - 1) z_1}{1 + (k_1 - 1)\frac{V}{F}} + \frac{(k_2 - 1) z_2}{1 + (k_2 - 1)\frac{V}{F}} + \frac{(k_3 - 1) z_3}{1 + (k_3 - 1)\frac{V}{F}} = 0$$

$$\left\{ \frac{(5-1) z_1}{1 + (5-1)\frac{V}{F}} + \frac{(1.9-1)\,0.35}{1 + (1.9-1)\frac{V}{F}} + \frac{(0.3-1)(0.65 - z_1)}{1 + (0.3-1)\frac{V}{F}} = 0 \right\}$$ equation #2

Two equations & two unknowns

	A	B	C	D	E	F	G	H	I
1	Problem 2.D10								
2	Components in mixture:	Propane	N-Butane	N-Hexane		Help			
3	K values:	5	1.9	0.3			Changing Values		
4	Mole Fraction:	0.23756	0.35	0.4124413			Given		
5	V/F:	0.58685					Based of calculation references		
6									
7									
8	Equations:	LHS	RHS	Equation Number					
9	Rachford-Rice Equation:	2E-14	0	#1					
10	Liquid Mole Fraction Hexane	0.7	0.7	#2					

Solution



PARAMETERS USED

$\frac{V}{F} = 0.587$
$Z_{C6} = 0.412$

5) PROBLEM 2.H6

Solve example 2-2 w/ a spreadsheet and Goal Seek using the k values from the example. Note that this spreadsheet is considerably simpler than the spreadsheets shown in Figures 2-B3 and 2-B4 in this Chapter's Appendix B, section 2.B2, because K values are not calculated in the spreadsheet

EXAMPLE 2-2

A flash chamber operating at 50°C and 200.0 kPa is seperating 1000.0 kmol/h of a feed that is 30.0 mol% propane, 10.0 mol% n-butane, 15.0 mol% n-pentane, and 45.0 mol% n-hexane. Find the product compositions and flow rates.

#1 Propane : $k_1 = 7.0$
#2 n-Butane : $k_2 = 2.4$
#3 n-Pentane: $k_3 = 0.80$
#4 n-Hexane: $k_4 = 0.30$

~~Equations~~

$$X_i = \frac{z_i}{1+(k_i-1)\frac{V}{F}}$$

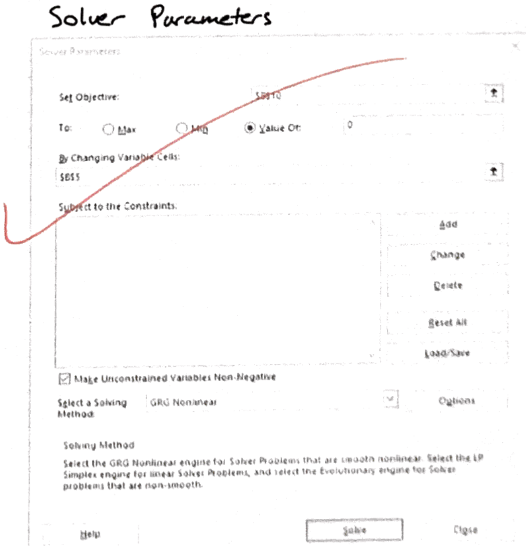
Rachford-Rice

$$\sum \frac{(k_i-1)z_i}{1+(k_i-1)V/F} = 0$$

Initial w/ guess

	A	B	C	D	E
1	Problem 2.H6				
2	Components in Mixture:	Propane	N-Butane	N-Pentane	N-Hexane
3	K Values:	7	2.4	0.8	0.3
4	Molar Composition:	0.3	0.1	0.15	0.45
5	V/F:	0.5			
6	Liquid Composition:	0.075	0.058824	0.1666667	0.692308
7	Vapor Composition:	0.525	0.141176	0.1333333	0.207692
8					
9	Equations	LHS	RHS		
10	Rachford-Rice Equation:	0.0144	0		

Solver Parameters



Solution

	A	B	C	D	E
1	Problem 2.H6				
2	Components in Mixture:	Propane	N-Butane	N-Pentane	N-Hexane
3	K Values:	7	2.4	0.8	0.3
4	Molar Composition:	0.3	0.1	0.15	0.45
5	V/F:	0.51137			
6	Liquid Composition:	0.07374	0.058278	0.1670889	0.700891
7	Vapor Composition:	0.51619	0.139867	0.1336711	0.210267
8					
9	Equations	LHS	RHS		
10	Rachford-Rice Equation:	-2.8E-15	0		
11					

$\frac{V}{F} = 0.51$ & $F = 1000.0$ kmol/h

$$\boxed{\begin{aligned} V &= 511.37 \text{ kmol/h} \\ L &= 488.63 \text{ kmol/h} \\ X_1 &= 0.0737 \\ X_2 &= 0.0583 \\ X_3 &= 0.1671 \\ X_4 &= 0.7009 \\ Y_1 &= 0.5162 \\ Y_2 &= 0.1399 \\ Y_3 &= 0.1337 \\ Y_4 &= 0.2103 \end{aligned}}$$

1 → Propane
2 → n-Butane
3 → n-Pentane
4 → n-Hexane